

Vishal Sreevidya Rajan

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PROFESSIONAL SUMMARY

Senior aerospace engineering student with hands-on experience in manufacturing research, multidisciplinary aircraft design, materials testing, machining, and engineering analysis. Skilled in MATLAB, SOLIDWORKS, CNC machining, Lean Six Sigma, and project management, with a focus on manufacturing, process, and quality engineering in aerospace and advanced manufacturing.

ENGINEERING EXPERIENCE

Manufacturing Research Assistant | UT Knoxville | June 2026 - Present

- Conducted hot-compression experiments on 316L stainless-steel cylindrical and double-cone specimens using a Gleeble system analyzing load, temperature, and deformation behavior across 200-1,400°C in QuikSim
- Designed and fabricated custom 3D-printed components in SOLIDWORKS to support graduate research experiments and streamline test setup and execution
- Operated a Model 4660 HiTorque Mill to machine and level 304L stainless-steel surfaces in support of graduate research experiments

PROJECTS

Sport Jet Aircraft Design | Aerospace Design Project | August 2026 - Present

- Collaborating on the preliminary design of an affordable two-seat sport jet targeting Mach 0.78, a +6g sport-flight capability, and a 600-nautical-mile ferry mission
- Supporting multidisciplinary aircraft design and trade studies involving aerodynamics, structures, propulsion, weight and balance, stability and control, systems integration, and mission performance

Hypersonic Glide Vehicle Design | JHTO-UCAH Competition | August 2026 - Present

- Contributing to the design of an unpowered hypersonic glide vehicle for controlled low-altitude hypersonic flight as a member of the Structures and Trajectory teams
- Evaluating structural and material requirements while supporting trajectory analysis that connects flight conditions with aerodynamic loads, thermal constraints, and vehicle-design trade-offs using Ansys and SOLIDWORKS

EcoCAR Integrated Vehicle Architecture | Vehicle Architecture Team | September 2026 - Present

- Defining a preliminary eAWD architecture that integrates the stock front electrified powertrain with a single-motor electric rear-axle drive (ERAD) feeding the rear differential
- Identifying powertrain components, battery power and energy demands, and off-road performance metrics required by cross-functional teams to establish traceable targets and guide architecture trade studies

LEADERSHIP EXPERIENCE

Ambassador | Anderson Center for Entrepreneurship & Innovation | February 2025 - Present

- Delivered entrepreneurship presentations to groups of 30+ high school students, explaining core concepts and early-stage venture development
- Collaborated with fellow ambassadors to develop outreach strategies that promoted Anderson Center programs and events

SKILLS

Manufacturing: CNC Machining, Milling, Additive Manufacturing

Software: MATLAB, SOLIDWORKS, Fusion 360, Mastercam, Microsoft Office Suite, Ansys

Professional: Technical Documentation, Team Collaboration, Attention to Detail

EDUCATION

University of Tennessee, Knoxville

May 2027

B.S. Aerospace Engineering | Minor in Engineering Management

CERTIFICATIONS

Lean Six Sigma Green Belt

Certified Associate in Project Management (CAPM)